Folding in R

Origami, Manifolds, and Dimension Reduction

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Contents

Preface

A rigid origami crease pattern generates a manifold whose true low-dimensional structure is known exactly, in closed form. Rigid folding bends the sheet along creases only; the facets stay flat. The folded surface is therefore isometric to the flat sheet, and geodesic distances measured along the surface are unchanged by folding. The flat crease pattern is not an approximation of the correct two-dimensional embedding. It *is* the correct embedding — the answer key.

That single property is what this book is built on.

Standard benchmarks for dimension reduction — the Swiss roll, the S-curve, the severed sphere — give one fixed geometry and a visual impression of success. A crease pattern gives an exact reconstruction error and a parameterised family of test problems. The fold angle θ runs continuously from a flat sheet to a fully compressed one, so every method's performance becomes a curve rather than a single number, and no configuration can be cherry-picked.

Chari and Pachter have argued that two-dimensional embeddings of high-dimensional biological data are systematically misleading. The usual rebuttal is anecdotal. On a manifold where the truth is known exactly, the distortion can be *measured* instead of argued about — and, more usefully, the evaluation metrics themselves can be checked against truth. That is the reason this book exists.

What this book is not

It is not a treatment of rigidity theory, computational origami design, or deep generative modelling. It is not a tutorial on `bookdown`, `make`, or project tooling. Where a section could not be tied back to the ground-truth claim above, it was cut rather than kept for completeness.

Impressum

Folding in R: Origami, Manifolds, and Dimension Reduction

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Source repository: https://github.com/r-heller/folding-in-r
Online edition: https://r-heller.github.io/folding-in-r/

This is a living document. The version you are reading was rendered on 2026-07-29.

Acknowledgments

This book stands on two bodies of work that developed independently and rarely cite each other: the mathematics of rigid origami, and the practice of non-linear dimension reduction. Bringing them together is the only original move here; both halves were built by others.

The R package ecosystem this book depends on — `bookdown`, `Rtsne`, `umap`, `igraph`, `vegan`, `torch` and their dependencies — is maintained largely by volunteers. The bibliography in `packages.bib` is generated automatically so that those authors are cited by name rather than thanked in the abstract.

How to use this book

Standing methodological rules

Three rules apply throughout and are not restated in each chapter.

1. **Every stochastic result is reported across at least 20 seeds.** t-SNE and UMAP are seed-dependent; a single run is an anecdote. Seeds are fixed in `_common.R` as `BENCH_SEEDS` and disclosed with every figure.
2. **Every θ-dependent result is reported as a curve, never as a single number.** Fold angle is a continuous difficulty parameter. Reporting one value invites cherry-picking, and hides exactly the regime changes the benchmark exists to expose.
3. **Every figure carries `fig.alt`.** No result is encoded by colour alone.

What you need

R 4.1 or newer and the packages pinned in `renv.lock`. Run `renv::restore()` once after cloning. The companion package `foldbench` provides the pattern constructors, the sampler, and the evaluation metrics; the book loads it but does not reproduce its source.

The Chapter 10 benchmark grid takes hours to compute. It is precomputed and committed under `data/processed/`; `scripts/run-benchmark-grid.R` is kept as the provenance record rather than run during the build.

How to read it

Part I builds the object. Part II tests four families of method against it. Part III is the benchmark proper and contains the book's main contribution — Chapter 9, where the evaluation metrics themselves are checked against exact truth. Part IV asks whether any of it changes a real decision.

Readers who only want the benchmark can start at Chapter 8 and refer back to Chapter 3 for the generative model.

Part I

Folding

Chapter 1

Introduction

Why should a crease pattern tell you anything about a gene expression matrix?

- The ground-truth thesis: a rigid crease pattern *is* the answer key, not a proxy for it.
- What the book claims, and what it explicitly does not claim.
- The Chari & Pachter challenge to 2D embeddings, stated up front as the target.
- Why exact truth lets us evaluate the evaluation metrics, not just the methods.
- Roadmap: fold the sheet (Part I), test the methods (II), build the benchmark (III), change a decision (IV).

Chapter 2

The geometry of folding

What makes a crease pattern a valid manifold?

- Developability and why a folded sheet has zero Gaussian curvature on facets.
- Rigid vs. non-rigid folding; why only rigid folding preserves the isometry we rely on.
- Maekawa's theorem: $|M - V| = 2$ at every interior vertex of a flat-foldable pattern.
- Kawasaki's theorem: alternating angle sums at an interior vertex are equal.
- Boxed aside, one page, labelled background: the Huzita–Hatori axioms.
- Piecewise-flat surfaces sit *outside* the smoothness assumptions of manifold-learning theory. Stated as realism, not as a caveat — real data manifolds have creases too.

Chapter 3

Folding as a generative model

How do you turn one pattern into a family of test problems?

- The Miura-ori parameterisation (a, b, α, θ).
- Uniform sampling on facets, and why naive vertex sampling biases the density.
- Noise models: ambient Gaussian, on-manifold, and outliers.
- What θ does to the k-nearest-neighbour graph, and where short-circuits appear.
- Why every result downstream is a curve over θ rather than a single number.

Part II

Methods under test

Chapter 4

Linear projections and where they break

When is a projection just a shadow?

- PCA via SVD with `prcomp`; why not `princomp` or an eigendecomposition of the covariance.
- At $\theta = 0$ two components are exact. Variance explained collapses as θ rises.
- One curve — and it motivates the whole book.
- Kernel PCA as the bridge into Part II.

Chapter 5

Geodesic methods

If the data lie on a folded sheet, how do you flatten it back out?

- Classical MDS with `cmdscale` as the baseline.
- Isomap: geodesics estimated by shortest paths on a kNN graph (`igraph::distances`).
- Locally linear embedding, and what it assumes about local neighbourhoods.
- Isomap is *built* for isometric embeddings, so the interesting result is where it breaks.
- Report the exact θ at which the kNN graph first bridges two facets — analytically and empirically.

Chapter 6

Neighbour embeddings

Why do t-SNE and UMAP disagree about the same data?

- `Rtsne` with `check_duplicates = TRUE`; document the duplicate-row failure mode.
- `umap(method = "naive")` — the default backend pulls Python `umap-learn`.
- Perplexity and `n_neighbors` sweeps.
- All results across `BENCH_SEEDS`, reported as distributions rather than means.
- Separate real disagreement from seed noise.

Chapter 7

Learned folds

Can a network learn the fold and the unfold at once?

- A `torch` autoencoder with a two-unit bottleneck. (`keras` would pull TensorFlow and break clean-clone reproducibility.)
- The only method here that learns an explicit inverse: encoder = fold, decoder = unfold.
- Reconstruction loss is the fold–unfold identity made computable.
- Compare the learned decoder against the true unfolding, not just against the input.

Part III

The benchmark

Chapter 8

Building crease-pattern manifolds

How do you build a manifold whose true geometry you already know?

- The `foldbench` API end to end.
- Pattern families beyond Miura-ori: Yoshimura, waterbomb.
- Product constructions: intrinsic dimension $2k$ in ambient $3k$, ground truth still exact.
- Random linear lift to arbitrary ambient dimension D.
- Why a benchmark stuck at intrinsic dimension 2 is not a serious benchmark.

Chapter 9

Ground truth and the evaluators

Your embedding looks good. Is it correct?

- Procrustes-aligned RMSE against the true unfolding as the headline metric.
- Trustworthiness, continuity, kNN preservation, Kruskal stress — each measured *against* truth.
- Where each standard metric is optimistic, and by how much.
- The metrics are kNN-based and mutually correlated; exact truth is what makes the comparison possible.
- The book's main experimental contribution.

Chapter 10

Benchmark results

Which method survives?

- The full method $\times$ pattern $\times$ θ $\times$ noise grid, at least 20 seeds per cell.
- Distributions, not means.
- State explicitly how much between-method variation falls inside within-method seed variance.
- Where Isomap wins, where it breaks, and the crossover.
- Grid is precomputed by `scripts/run-benchmark-grid.R` and committed; the script is the provenance record.

Chapter 11

Crease patterns vs. the Swiss roll

Does the new benchmark actually add anything?

- Head-to-head against Swiss roll and S-curve, same methods and same metrics.
- Method rankings that flip between benchmarks — and why.
- What the crease-pattern family surfaces that a single fixed geometry hides.
- An honest answer to “why not just use the Swiss roll?”

Part IV

Application

Chapter 12

From benchmark to decision

What happens when the data are real?

- Use the Chapter 10 grid to select a method and its hyperparameters.
- Apply that choice to a single-cell dataset.
- Compare against the field-default choice.
- Report a null result honestly if the benchmark-guided choice does not win.
- Closes with visualisation and honest-reporting standards: alt text, no colour-only encoding, seed disclosure, aspect-ratio discipline.

Appendix A

Notation

Symbol	Meaning
θ	Fold angle; the difficulty parameter
a, b, α	Miura-ori cell parameters
M, V	Mountain and valley creases
k	Neighbourhood size for kNN-graph methods
D	Ambient dimension after linear lift
$\mathcal{M}$	The folded manifold in $\mathbb{R}^3$
$\mathcal{U}$	The exact unfolding — ground truth in $\mathbb{R}^2$

Appendix B

Datasets and codebook

Generated data

All crease-pattern manifolds are generated, not collected. Every dataset in Parts I–III is reproducible from a pattern specification, a fold angle, a sample size, a noise specification, and a seed.

Precomputed results

`data/processed/benchmark-grid.rds` — the Chapter 10 grid. Provenance: `scripts/run-benchmark-grid.R`.

External data

Glossary

Developable surface A surface that can be flattened into a plane without stretching or tearing.

Fold angle (θ) The dihedral angle at a crease. Runs from 0 (flat sheet) to $\theta_{\max}$ (fully compressed) and serves as the difficulty parameter throughout.

Isometry A map that preserves distances measured *along* the surface. Rigid folding is an isometry; this is the property the whole book rests on.

Mountain / valley assignment The direction of each crease. Written M and V; encoded by linetype as well as colour in every figure.

Rigid origami Folding in which facets stay flat and all deformation happens at creases.

Short-circuit Two points close in ambient space but far apart geodesically. The failure mode that breaks kNN-graph methods.

Colophon

Written in R Markdown and built with **bookdown** (Xie, 2026). Typeset in Inter and JetBrains Mono. Figures use the viridis plasma palette (option `"C"`), the series convention for this volume.

Interface icons are Font Awesome Free 6.5.1, vendored in `style/webfonts/` rather than loaded from a content delivery network. The icons are licensed CC BY 4.0, the fonts under the SIL OFL 1.1, and the accompanying code under the MIT License; see https://fontawesome.com/license/free.

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#>  quarto   1.9.38 @ /usr/local/bin/quarto
#>
#> ------------------------------------------------------------------------------
```

About the author

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References

Bibliography

Xie, Y. (2026). *bookdown: Authoring Books and Technical Documents with R Markdown*. R package version 0.47.